

EOX Multiple Reactions in a PFR

The following reactions are carried out in the gas phase in a 400-dm³ PFR at 325 K:

3A → B + C	-r _{1A} = k _{1A} C _A	k _{1A} = 7.0 min ⁻¹
2C + A → 3D	r _{2D} = k _{2D} C _C ² C _A	k _{2D} = 3.0 dm ⁶ /mol ² *min
4D + 3C → 3E	r _{3E} = k _{3E} C _C C _D	k _{3E} = 2.0 dm ³ /mol*min

$$\frac{r_{1A}}{-3} = \frac{r_{1B}}{1} \quad \text{so} \quad r_{1B} = -\frac{1}{3} r_{1A}$$

A is fed at a concentration of 0.2 mol/dm³, and the inlet volumetric flowrate is 100 dm³/min. Plot the concentrations of A, B, C, D, and E in the reactor and determine $\bar{S}_{D/E}$ and $\bar{S}_{C/D}$.

mole balances

$$\frac{dF_A}{dV} = r_A \quad \frac{dF_B}{dV} = r_B \quad \frac{dF_C}{dV} = r_C$$

$$\frac{dF_D}{dV} = r_D \quad \frac{dF_E}{dV} = r_E$$

net rates of generation

$$r_A = r_{1A} + r_{2A}$$

$$r_B = r_{1B}$$

$$r_C = r_{1C} + r_{2C} + r_{3C}$$

$$r_D = r_{2D} + r_{3D}$$

$$r_E = r_{3E}$$

relative rates

$$r_{1B} = -\frac{1}{3} r_{1A}$$

$$r_{1C} = r_{1B}$$

$$r_{2C} = -\frac{2}{3} r_{2D}$$

$$r_{2A} = -\frac{1}{3} r_{2D}$$

$$r_{3D} = -\frac{4}{3} r_{3E}$$

$$r_{3C} = -r_{3E}$$

concentration equations $\overset{0}{\rightarrow}$ $\overset{=1, \text{ isothermal}}{\rightarrow}$

$$C_j = C_{T0} \left(\frac{F_j}{F_T} \right) \left(\frac{P}{P_0} \right) \left(\frac{T}{T_0} \right)$$

$\downarrow$
 $\downarrow$
 $\downarrow$
 $=1, \text{ no pressure drop}$

$$F_T = F_A + F_B + F_C + F_D + F_E$$

$$C_A = C_{T0} \frac{F_A}{F_T} \quad C_B = C_{T0} \frac{F_B}{F_T} \quad C_C = \frac{F_C}{F_T} C_{T0}$$

$$C_D = C_{T0} \frac{F_D}{F_T} \quad C_E = C_{T0} \frac{F_E}{F_T}$$